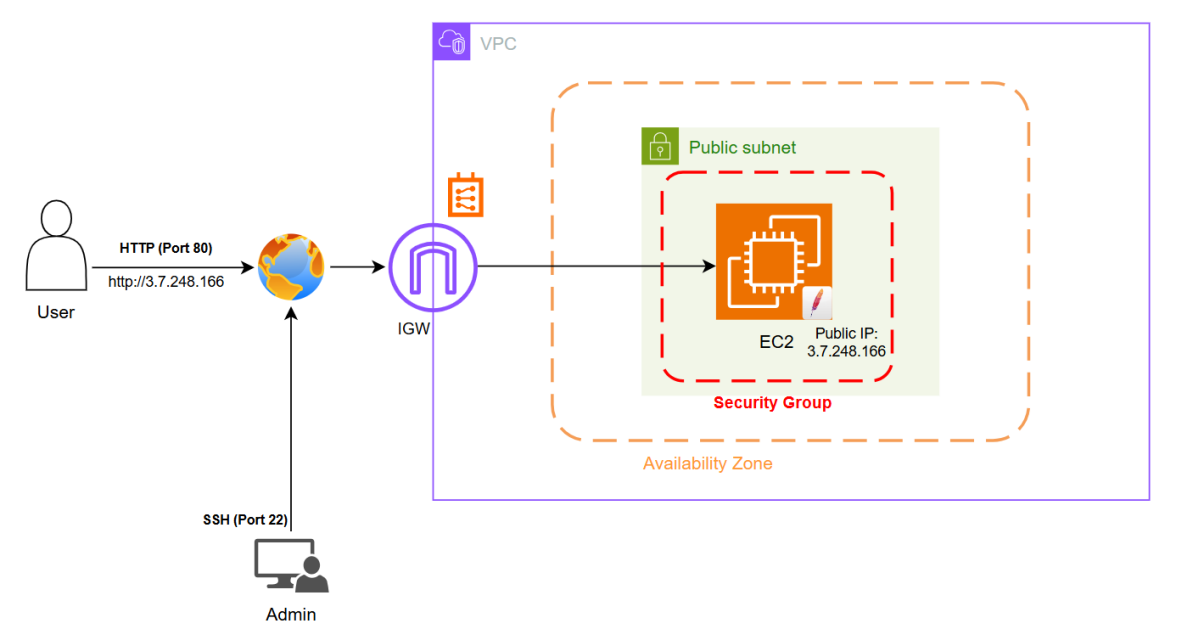
Project : EC2-based Web Server – Configure Apache on an EC2 instance.

In this project, I will deploy a web server using Amazon EC2. I will launch an EC2 instance, install Apache, and configure it to serve a basic website.



AWS EC2 Apache Web Server Architecture

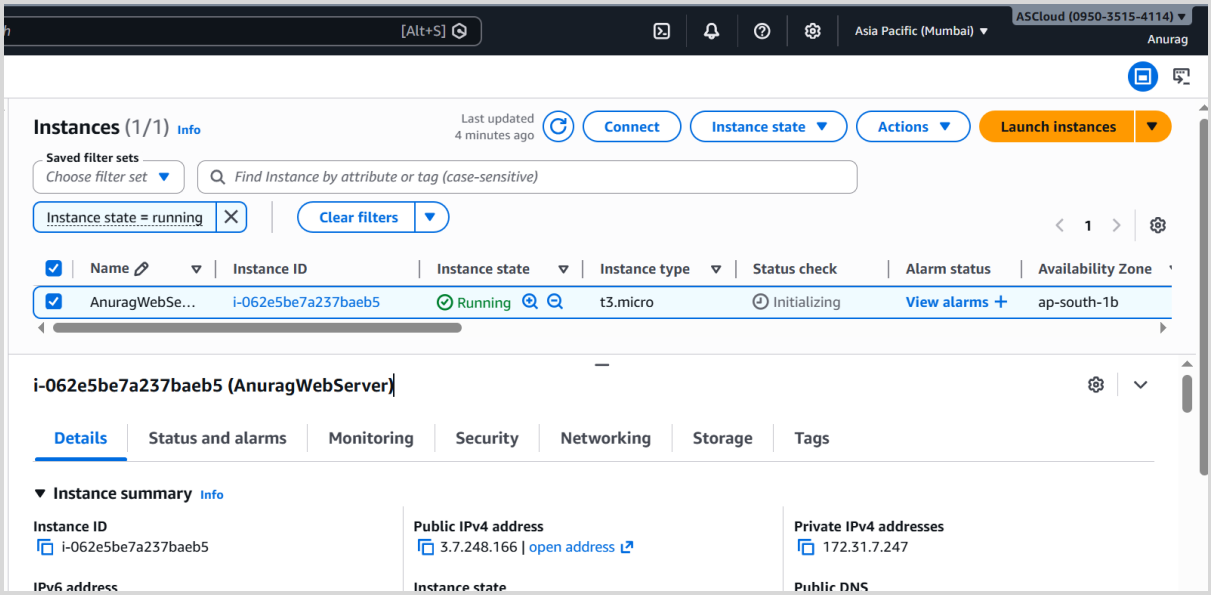
Step 1 — Launch an EC2 Instance

1. Go to **AWS Console → EC2 → Launch Instance**

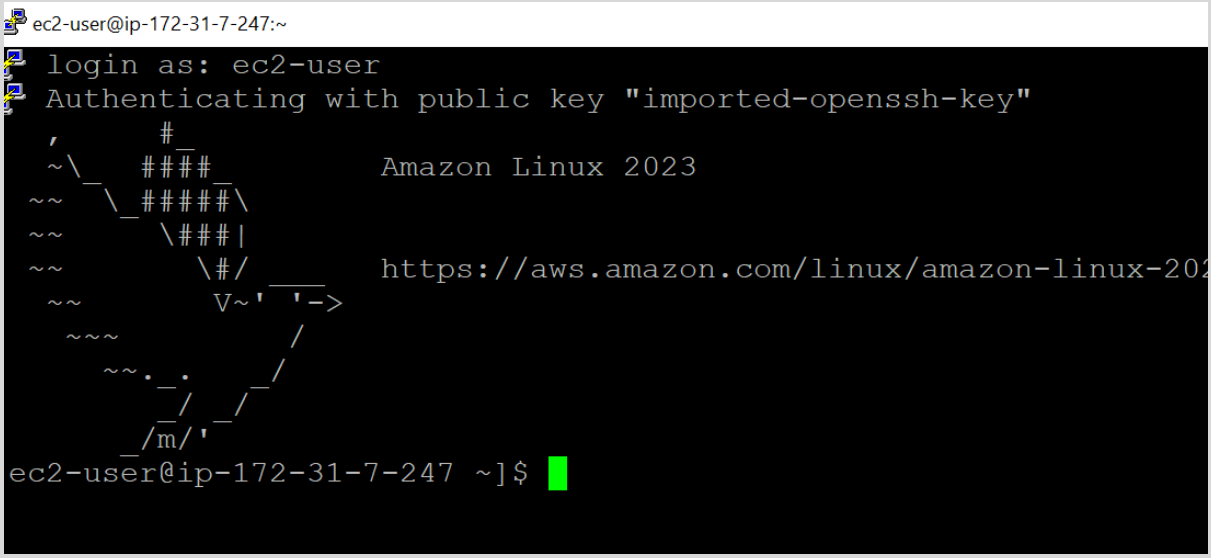
Configure:

- **Name:** AnuragWebServer
- **AMI:** Amazon Linux 2023
- **Instance type:** t3.micro
- **Key pair:** Create new key pair (.pem)
- **Network settings**
  - VPC: Default
  - Subnet: Public subnet
  - Auto-assign Public IP: **Enable**
- **Security Group:** Allow SSH(MY IP) and HTTP(Anywhere)

Click Launch



Step 2 — Connect to the EC2 Insatnce



Step 3 — Update System

Always update first.

sudo yum update –y

```
[ec2-user@ip-172-31-7-247 ~]$ sudo yum update -y
```

Step 4 — Install Apache Web Server

Install Apache:

sudo yum install httpd –y

```
  Verifying        : apr-1.7.5-1.amzn2023.0.4.x86_64                      1/
  Verifying        : apr-util-1.6.3-1.amzn2023.0.2.x86_64                 2/
  Verifying        : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64            3/
  Verifying        : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64         4/
  Verifying        : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch    5/
  Verifying        : httpd-2.4.66-1.amzn2023.0.1.x86_64                   6/
  Verifying        : httpd-core-2.4.66-1.amzn2023.0.1.x86_64              7/
  Verifying        : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch        8/
  Verifying        : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64             9/
  Verifying        : libbrotli-1.0.9-4.amzn2023.0.2.x86_64               10/
  Verifying        : mailcap-2.1.49-3.amzn2023.0.3.noarch                11/
  Verifying        : mod_http2-2.0.27-1.amzn2023.0.3.x86_64              12/
  Verifying        : mod_lua-2.4.66-1.amzn2023.0.1.x86_64                13/

Installed:
  apr-1.7.5-1.amzn2023.0.4.x86_64
  apr-util-1.6.3-1.amzn2023.0.2.x86_64
  apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64
  apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
  httpd-2.4.66-1.amzn2023.0.1.x86_64
  httpd-core-2.4.66-1.amzn2023.0.1.x86_64
  httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch
  httpd-tools-2.4.66-1.amzn2023.0.1.x86_64
  libbrotli-1.0.9-4.amzn2023.0.2.x86_64
  mailcap-2.1.49-3.amzn2023.0.3.noarch
  mod_http2-2.0.27-1.amzn2023.0.3.x86_64
  mod_lua-2.4.66-1.amzn2023.0.1.x86_64

Complete!
```

Check version

httpd –v

```
[ec2-user@ip-172-31-7-247 ~]$ httpd -v
Server version: Apache/2.4.66 (Amazon Linux)
Server built:   Dec  5 2025 00:00:00
[ec2-user@ip-172-31-7-247 ~]$
```

Step 5 — Start and enable Apache Service

Start the web server:

sudo systemctl start httpd

```
[ec2-user@ip-172-31-7-247 ~]$ sudo systemctl start httpd
[ec2-user@ip-172-31-7-247 ~]$
```

Enable auto start:

sudo systemctl enable httpd

```
[ec2-user@ip-172-31-7-247 ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.servi
ce.
```

Check status:

sudo systemctl status httpd

```
[ec2-user@ip-172-31-7-247 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
     Active: active (running) since Mon 2026-04-20 09:56:15 UTC; 1min 40s ago
       Docs: man:httpd.service(8)
   Main PID: 25904 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 1014)
     Memory: 13.4M
        CPU: 171ms
     CGroup: /system.slice/httpd.service
             ├─25904 /usr/sbin/httpd -DFOREGROUND
             ├─25927 /usr/sbin/httpd -DFOREGROUND
             ├─25928 /usr/sbin/httpd -DFOREGROUND
             ├─25929 /usr/sbin/httpd -DFOREGROUND
             └─25930 /usr/sbin/httpd -DFOREGROUND
```

Step 6 — Deploy Web Page

Create a simple web page.

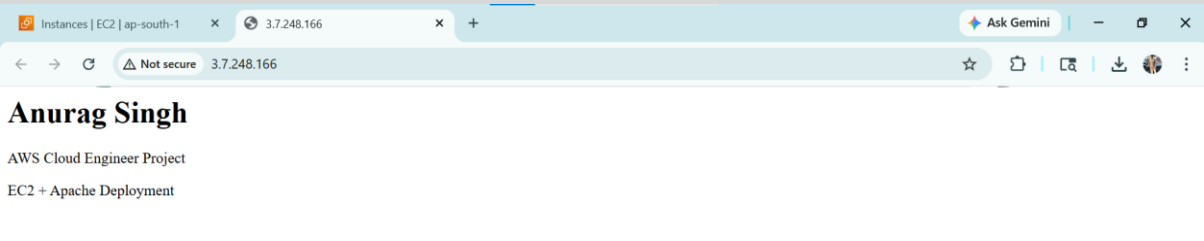
sudo nano /var/www/html/index.html

Add:
<h1>Anurag Singh</h1>
AWS Cloud Engineer Project
EC2 + Apache Deployment

Save it and exit

```
[ec2-user@ip-172-31-7-247 html]$ cat index.html
<h1>Anurag Singh</h1>
<p>AWS Cloud Engineer Project</p>
<p>EC2 + Apache Deployment</p>
```

Step 7 — Access Web Server

Open browser and go to:

http://3.7.248.166